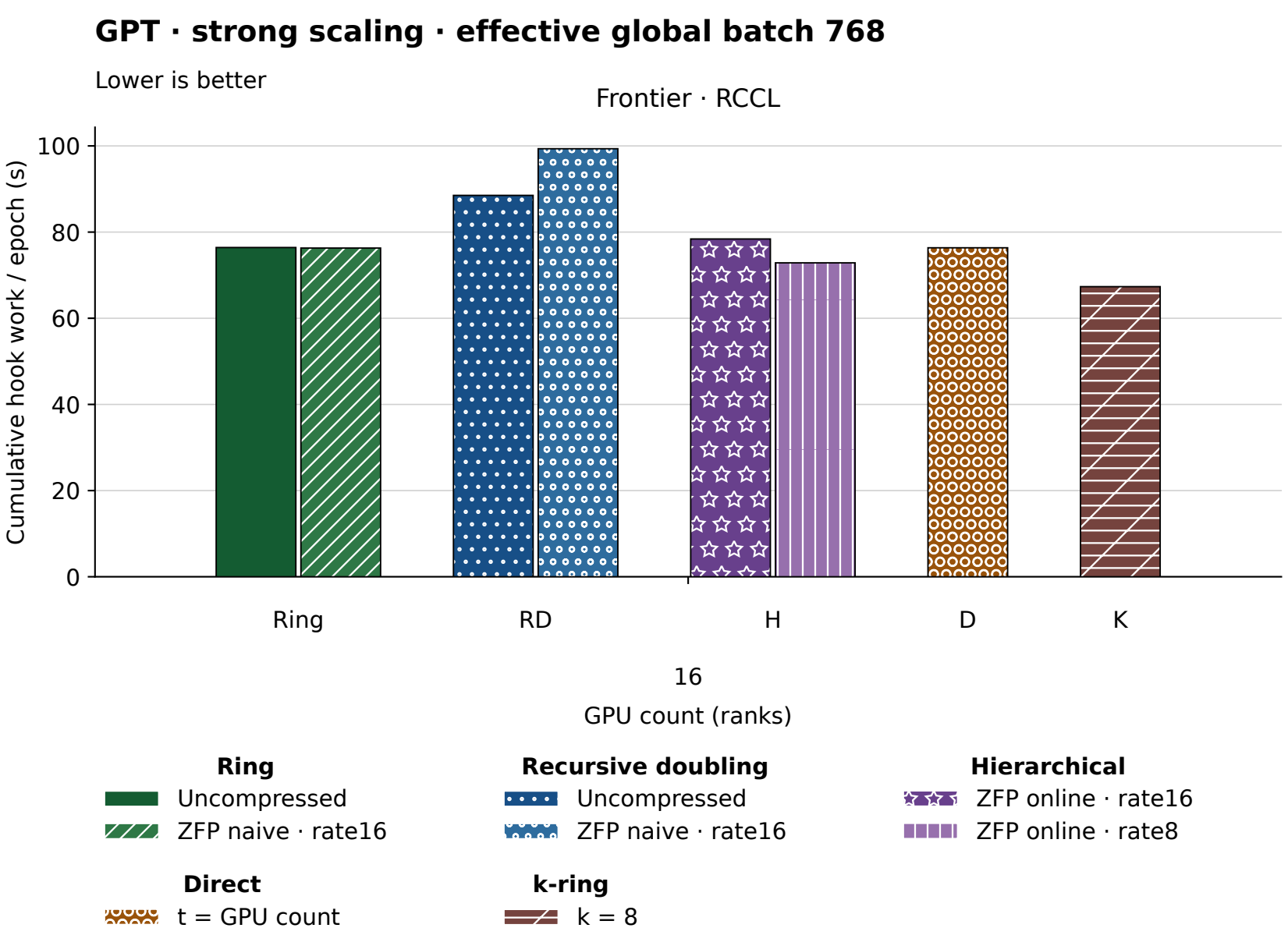

GPT · strong scaling · effective global batch 768
Lower is better
Frontier · RCCL
Cumulative hook work / epoch (s)
100
80
60
40
20
0
Ring
RD
H
D
K
16
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Direct
t = GPU count
k-ring
k = 8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate